THE PROBLEM

A GST officer can’t tell which factories are hiding production or inflating tax credits, because every document she checks is declared by the factory itself, while the independent evidence (power use, goods movement, supplier records, bank flows) sits in separate places and is never compared. If this worked, she would start each week with a short list of units whose paperwork contradicts physical reality, each contradiction spelled out, and honest-but-unusual factories cleared.

THE EVIDENCE GAP

Declarations are only one part of the story.

01

DECLARED

Invoices & returns

What the unit reports

02

PHYSICAL

Power & stock

What production could support

03

MOVEMENT

E-way bills

What actually left the gate

04

FINANCIAL

Bank flows

What payments show

05

NETWORK

Supplier returns

Whether the chain agrees



EVIDENCE RECONCILIATION

One factory · one period · linked proof

Each gap carries its source rows into review.

A contradiction becomes explainable when independent trails are compared.

THE METHOD

Four checks; one explainable review queue.

01

Material in → declared output

Purchased inputs vs output; power is corroborating evidence.

02

Dispatch → invoice

E-way bill quantity vs invoiced quantity; bank inflow is supporting evidence.

03

Stock roll-forward

Opening + purchases + production − dispatch = closing.

04

Supplier-chain trace

Related-party links, invoice loops and supplier return mismatches.

PRIORITISE WITH 2+ DISTINCT SOURCE FAMILIES

Single-source cases stay outside the priority queue

LLM CASE NOTE

Summarises evidence rows; never flags.

MEASURED PROTOTYPE

Held-out evaluation exposes the false-positive cost

84.42%

monthly cheat recall

3409 / 4038 active cheat months

16.76%

monthly false-positive rate

4151 / 24762 negative months

629

active cheat months missed

5 cheat factories never caught

HELD-OUT SEED B

2,400 factories

28,800 factory-months

228,854 raw evidence rows

ANNUAL SCORES AND QUEUE PRECISION

Factory recall	**99.31%**
Factory false-positive rate	**84.17%**
Monthly queue precision @ 50	**18.17%**
Monthly queue precision @ 200	**51.58%**
Single-signal decoy months	**6924**

Synthetic data only. Threshold selected on A. High FPR prevents operational use. CIs and misses in README.

A RELATED PATTERN IN GERMANY

Invoice chains matter across borders.

A possible adaptation requiring local evidence and validation



A Germany adaptation would reconcile invoice chains, goods movements and payments under locally validated rules.

This prototype demonstrates the evidence-reconciliation pattern; it does not implement German VAT rules.

PATTERN COMPARISON